

04.2 RG Intervention Framework Codebook

Field definitions, controlled vocabularies, and decision rules for the A Safe Bet Responsible Gambling Intervention Framework (210 tools)

Companion to: 04.1 RG Intervention Framework (04_1_RG_Intervention_Framework.xlsx)

1. Overview

1.1 Purpose

The RG Intervention Framework (file 04.1) is the second-stage output of the A Safe Bet review programme. It consolidates the 410 tool records produced in Stage 1 into 210 distinct responsible gambling (RG) tool concepts, each described against a standardised structure of fifteen domains and six categorical dimensions.

This codebook defines every column in the Framework, the controlled vocabularies used to populate them, and the decision rules applied during coding. It is the reference document for anyone reading, citing, extending, or operationalising the Framework.

1.2 Relationship to Stage 1

Stage 1 produced the systematic review's coded evidence base: 410 tool records extracted across a 45-field codebook, with full search, screening, and quality-rating documentation. Stage 2 takes that evidence base and consolidates it: where multiple studies test functionally identical tools, those records collapse into a single tool entry; where reviewed evidence implies tools that do not yet exist as discrete interventions, those tools are added as theoretically derived or extrapolated.

This codebook does not duplicate Stage 1 documentation. Where upstream methods are relevant (search strategy, eligibility criteria, extraction codebook, evidence-rating rubric), readers are directed to:

- 01.1 Search Strategy
- 01.2 Eligibility Criteria
- 01.4 PRISMA Flow
- 02.2 Extraction Codebook
- 02.3 Tool Codes and Domains
- 02.5 Evidence Quality and Readiness

1.3 File structure

The deposited file is 04_1_RG_Intervention_Framework.xlsx. It contains one row per tool, organised by domain, across three sheets described in Section 2.3.

1.4 How to read this codebook

Section 2 describes the Framework's macro-architecture: the fifteen domains, the tool ID convention, the three sheets in the workbook, and the visual conventions used in the file.

Section 3 documents each of the thirteen columns. For each column, the codebook gives: purpose, format, controlled vocabulary (where applicable), and the decision rules applied during coding. Decision rules are embedded in the relevant column section rather than in a separate decisions chapter.

Section 4 documents the six columns of the workbook's reference sheets (Domain Summary and Tool Code Reference).

Counts and statistics are not included in this codebook. The codebook documents structure, vocabularies, and decision rules; quantitative summaries (record counts, distributions, cross-tabulations) appear in the main paper.

2. Framework architecture

2.1 Tool ID convention

Each tool has a unique identifier of the form T#### where ### is a zero-padded three-digit integer from 001 to 210. Tool IDs are stable: once assigned, they are never reused or renumbered. New tools added to future versions of the Framework extend the numbering rather than reorganise it.

The numbering reflects the order in which tools were added during the consolidation work, not their conceptual rank. Tools T001 through T174 represent the consolidated empirical and theoretical evidence base. Tools T175 through T210 were added in a final extrapolation pass to fill identified architectural gaps; most carry Evidence Status = Extrapolated, although a small number (T200, T207) gained empirical support during the trace audit and now carry Evidence Status = Empirically tested while remaining in the extrapolated numbering block by ID position.

2.2 Domains and section headers

Tools are grouped into fifteen domains, each occupying a contiguous block of Tool IDs. The file uses coloured section header rows to mark domain boundaries, with each domain assigned a distinct hue. The fifteen domains are:

- Financial Consumption Limits (LIMIT-DEP, LIMIT-LOSS, LIMIT-WIN, LIMIT-BET)
- Time and Session Management (LIMIT-TIME, BREAK-FORCE)
- Exclusion and Account Controls (EXCL-TEMP, EXCL-LONG)
- Messaging and Communications (MSG-STATIC, MSG-POP, MSG-PUSH)
- Personalised Feedback (FEED-TRACK, FEED-NORM, FEED-PERSONAL, FEED-RISK)
- Detection and Monitoring Systems (DETECT)
- Self-Assessment Tools (ASSESS-SELF)
- Education and Information (EDU-INFO, EDU-INTERACT, EDU-VIDEO)
- Proactive Contact (CONTACT-PROACTIVE)
- Therapeutic Interventions (THER-CBT, THER-MI, THER-SELF, THER-COMBO)
- Self-Regulation Strategies (SELF-REG)
- Structural Game Modifications (STRUCT-DISPLAY, STRUCT-OTHER, STRUCT-PAY, STRUCT-STAKE, STRUCT-TEMPO)
- Low-Risk Gambling Guidelines (GUIDE-LRGL)
- Multi-Component Programmes and System Tools (PROGRAM)
- Extrapolated and Novel Tools (mixed categories)

2.3 Workbook sheets

The workbook contains three sheets:

RG Intervention Framework. The main data sheet containing one row per tool, organised by domain. This is the primary artefact described by Section 3 of this codebook.

Domain Summary. Reference sheet listing each domain with its constituent tool codes and a short description. Provides navigation and orientation.

Tool Code Reference. Reference sheet listing each tool code with its full name and parent domain. Provides decoding for shorthand codes used in the Framework.

Counts are not held on the reference sheets. The codebook documents structure only; quantitative summaries appear in the main paper.

2.4 Visual conventions

The file uses the following visual conventions for orientation:

Domain section headers. Dark coloured rows with white bold text marking the start of each domain block. Each domain is assigned its own hue; the same hue is used (in lighter tints) for the rows below.

Domain-tinted zebra striping. Tool rows within each domain alternate between two pale tints of the domain's hue. This makes the domain boundaries visible while keeping the table readable.

Evidence Status colour coding. The Evidence Status cell in each row is filled according to its value: pale green for Empirically tested, pale yellow for Theoretically derived, pale blue for Policy / structural, pale orange for Extrapolated.

Header row. The column header row is dark navy with white bold text and is frozen so that headers remain visible during scrolling.

3. Column definitions and rules

Each subsection documents one column of the Framework. Columns are presented in the order they appear in the OSF clean file.

3.1 Tool ID

Purpose	Unique stable identifier for each tool.
Format	T### (e.g., T042). Zero-padded three-digit integer.
Range	T001 through T210.
Decision rules	IDs are assigned in the order tools were added during consolidation. Once assigned, an ID is never reused or reassigned. New tools in future versions extend the numbering rather than reorganise it.

3.2 Domain

Purpose	Indicates the high-level domain to which the tool belongs.
Format	Free-text domain label (one of the fifteen domains listed in Section 2.2).
Decision rules	Domain is determined by the tool's primary mechanism, not by its evidence status. Tools T200 and T207 sit in the Extrapolated and Novel Tools domain block by ID position but carry Evidence Status = Empirically tested; the domain label reflects the tool's structural position in the Framework, not its current evidence base.

3.3 Tool Code

Purpose	Indicates the specific tool code to which the tool belongs. Tool codes are subdivisions of domains.
Format	Hyphenated uppercase code (e.g., LIMIT-DEP, MSG-POP, EXCL-TEMP).
Decision rules	Tool codes follow Stage 1's tool code system as documented in 02.3 Tool Codes and Domains. Where Stage 1 used compound codes (e.g., LIMIT-DEP + LIMIT-LOSS for

studies testing multiple limit types simultaneously), the Framework assigns the tool to its single primary code and uses the Taxonomy Source column to record the multi-code provenance.

3.4 Tool Name

Purpose	Human-readable name of the tool, sufficient to identify it without consulting the Tool Description.
Format	Title-case free text. May include a parenthetical qualifier indicating a key parameter (e.g., ‘60 min → 90-second lock’).
Decision rules	Names are written to be informative at a glance. They include the tool’s mechanism (e.g., ‘Mandatory Session Break’) plus a distinguishing parameter (e.g., ‘60 min continuous play → 90-second lock’) where multiple variants of the same mechanism exist. British spelling is used throughout. Em dashes are not used; en dashes (–) separate qualifiers.

3.5 Tool Description

Purpose	Concise description of the tool’s mechanism and parameters.
Format	Single paragraph, 60–100 words. Lead sentence states the tool’s mechanism; subsequent text gives parameters and design context.
Decision rules	Descriptions are written to support comparison across tools. They focus on what the tool does and how it operates, not on its evidence base or deployment specifics. British spelling. No em dashes. Active voice. Where parameters are key to distinguishing a tool from related variants, those parameters are stated explicitly (e.g., ‘mandatory 24-hour delay’; ‘cap at 1% of monthly income’).

3.6 Target Population

Purpose	Indicates the player population for which the tool is designed.
Format	Single value drawn from a controlled vocabulary.
Controlled vocabulary	Universal Moderate-risk Moderate-High-risk High-risk Young adults New accounts Research/system development
Decision rules	Coded by convergence of three criteria: (1) the mechanism’s design intent; (2) the friction profile (how restrictive the tool is); (3) the evidence base (which population the tool was tested on or designed for). Where these three criteria converge, code by their consensus. Where they diverge, code by the most restrictive population. For example, a tool whose mechanism implies universal applicability but whose evidence base tested only high-risk gamblers is coded as High-risk.

Note. The Target Population vocabulary is a Stage 2 elaboration of the Stage 1 record-level Tool_Target_Tier values (ALL, HIGH, MOD, LOW, N/A; see 02.1 Field Inventory and 02.2 Extraction Codebook). The Stage 2 vocabulary preserves the four risk-tier values (Universal, Moderate-risk, Moderate-High-risk, High-risk) and adds three further targeting categories (Young adults, New accounts, Research/system development) to capture targeting concepts that emerge at the framework-entry level but were not coded at the record level.

Value definitions

Universal. Designed for all players regardless of risk profile. Most voluntary RG tools fall here.

Moderate-risk. Targeted at players showing emerging risk markers but not yet meeting high-risk thresholds.

Moderate-High-risk. Targeted at players in the upper moderate or lower high-risk tiers; typical recipients of operator outreach interventions.

High-risk. Designed for players showing established risk indicators or in active treatment.

Young adults. Specifically targeted at the 18–25 age band, where lower-risk thresholds and elevated vulnerability are well-established.

New accounts. Designed for the registration window or first weeks of play, where behavioural data is limited but risk-marker emergence is rapid.

Research/system development. Tested or developed in a research or system-development context without a clear deployment population. Reserved for clearly methodological tools (e.g., decision-support paradigms tested only in laboratory conditions).

3.7 Delivery Context

Purpose	Indicates the channel and delivery surface through which the tool reaches the player.
Format	Compound: ‘Channel X; Delivery Y’ (e.g., ‘Channel Online; Delivery In-product’).
Channel vocabulary	Online Land-based Cross-channel Not channel-specific
Delivery vocabulary	In-product Clinical Community Laboratory Player-applied
Decision rules	Channel describes the gambling environment (online platform, land-based venue, cross-channel system, or channel-agnostic). Delivery describes the surface on which the tool is presented (within the product itself, in clinical settings, in community/educational settings, in laboratory paradigms, or applied by the player without operator infrastructure). Tools may be coded as ‘Cross-channel’ where the same mechanism is documented across both online and land-based contexts in the underlying evidence.

Channel value definitions

Online. Delivered through an online gambling platform (website or app).

Land-based. Delivered in a physical venue (casino, club, betting shop, EGM venue).

Cross-channel. Operates across both online and land-based contexts; typically implies a national-registry or cross-platform mechanism.

Not channel-specific. Channel-agnostic by design (e.g., self-regulation strategies the player applies regardless of context).

Delivery value definitions

In-product. Delivered within the gambling product or platform itself; the player encounters the tool while gambling or accessing their account.

Clinical. Delivered through a clinical or treatment context (face-to-face, telephone, or telehealth therapy; structured counselling).

Community. Delivered through community-facing channels (public messaging campaigns, educational materials, brochures, helplines accessed at point of need).

Laboratory. Delivered in research-environment paradigms (computerised tasks; experimental simulations not available outside the study).

Player-applied. The player implements the tool themselves without operator or clinician infrastructure (cognitive commitments, household-level cash management strategies).

3.8 Governance

Purpose	Indicates how the tool's adoption and operation are governed: who initiates use, and whether use can be reversed.
Format	Compound: 'Control X; Reversibility Y'
Control vocabulary	Mandatory Voluntary (opt-in) Voluntary (opt-out) Voluntary (self-initiated) Hybrid Operator-side
Reversibility vocabulary	Reversible Gated Irreversible N/A
Decision rules	Control captures who initiates and authorises the tool's operation. 'Mandatory' tools cannot be declined by the player. 'Voluntary (opt-in)' requires active player choice to enable; 'Voluntary (opt-out)' is enabled by default and requires player choice to disable. 'Voluntary (self-initiated)' captures tools the player applies independently without operator infrastructure. 'Hybrid' captures tools that combine player-side choice with operator-side enforcement. 'Operator-side' captures tools that operate entirely behind the scenes without player-facing controls (most detection systems). Reversibility captures whether the player can undo or modify their use of the tool: 'Reversible' immediately, 'Gated' through a delay or counselling requirement, 'Irreversible' permanently. 'N/A' applies where the concept does not apply (e.g., one-time messages, operator-side detection).

3.9 Personalisation

Purpose	Indicates whether and how the tool's content or parameters are tailored to the individual player.
Format	Single value (one of four base values) or a compound joined by '+'.
Base vocabulary	Non-tailored User-tailored System-tailored Practitioner-tailored
Decision rules	'Non-tailored' tools deliver identical content or parameters to all players. 'User-tailored' tools have content or parameters set by the player themselves. 'System-tailored' tools use algorithmic personalisation based on player data (behavioural risk scores, observed patterns). 'Practitioner-tailored' tools require actual judgement-based adaptation by a clinician, advisor, or staff member; mere delivery by a practitioner does not qualify. Compound values (joined by '+') indicate tools that combine multiple personalisation modes (e.g., 'User-tailored + System-tailored' for tools where the player sets a goal and the system tailors delivery).

Value definitions

Non-tailored. Identical content or parameters for all players who receive the tool.

User-tailored. The player sets their own parameters or selects from options; the system enforces the player's choice.

System-tailored. The system algorithmically tailors content or parameters based on observed player data without requiring player input.

Practitioner-tailored. A clinician, advisor, or trained staff member adapts the tool's content or delivery based on judgement-based assessment of the individual player. Pure delivery by a practitioner without judgement-based adaptation is not Practitioner-tailored.

3.10 Trigger

Purpose	Indicates what causes the tool to activate and at which stage of the player's gambling activity it operates.
Format	Compound: 'Trigger X; Stage Y'
Trigger vocabulary	User-initiated Automatic Operator-initiated Time-threshold Spend-threshold Behaviour-threshold Event-triggered Third-party-initiated N/A
Stage vocabulary	Pre-session During-play Post-session Between-sessions On-demand Scheduled Continuous N/A
Decision rules	Trigger captures what activates the tool. 'User-initiated' tools require active player action. 'Automatic' tools activate without any condition (e.g., always-on displays). 'Operator-initiated' tools are activated by operator decision (e.g., outreach calls). Threshold-based triggers ('Time', 'Spend', 'Behaviour') activate when the player crosses a defined parameter. 'Event-triggered' tools activate on specific events (e.g., wins, deposits). 'Third-party-initiated' covers tools triggered by family members or professionals. 'N/A' applies where the concept does not meaningfully apply (e.g., always-visible static signage). Stage captures when in the gambling cycle the tool operates.

3.11 Evidence Status

Purpose	Indicates the status of empirical evidence supporting the tool.
Format	Single value drawn from a controlled vocabulary.
Controlled vocabulary	Empirically tested Theoretically derived Policy / structural Extrapolated
Decision rules	Assigned by reference to the tool's Taxonomy Source. Tools with one or more substantive empirical traces are coded as Empirically tested. Tools with no empirical traces but conceptual grounding in the reviewed evidence are Theoretically derived. Tools that exist as regulatory or structural features (typically without RCT-level evaluation) are Policy / structural. Tools added by mechanistic extrapolation from the reviewed evidence base are Extrapolated.

Value definitions

Empirically tested. The tool, or a closely matched variant, has been directly evaluated in the reviewed literature.

Theoretically derived. The tool is conceptually motivated by the evidence base (mentioned, audited, or theorised) but has not been formally evaluated as a standalone intervention.

Policy / structural. The tool is a regulatory or structural feature; it is documented in the literature but typically without RCT-level evaluation. Used for policy-mandated controls (regulatory caps, mandatory signage, structural EGM modifications).

Extrapolated. Novel tool derived by mechanistic extrapolation from reviewed evidence. Suggested tool that warrants future evaluation; not yet tested in any form.

Colour coding in the file

The Evidence Status cell in each tool row is filled according to its value: pale green for Empirically tested, pale yellow for Theoretically derived, pale blue for Policy / structural, pale orange for Extrapolated.

3.12 Taxonomy Source

Purpose	Lists the Stage 1 tool records that substantively examine the tool, providing direct traceability between the Framework and the underlying evidence base.
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Format	Comma-separated list of Stage 1 study_id values, or empty for tools with no traceable evidence (typically Extrapolated tools).
Decision rules	Populated according to the substantive trace rule (described below). All 410 Stage 1 records are referenced at least once across the column; conversely, 169 of 210 tools have at least one trace, with the remaining 41 tools (extrapolated and proposed) having no trace.

3.12.1 The substantive trace rule

A Stage 1 record is included as a Taxonomy Source for a Framework tool if and only if the underlying study substantively examines the tool concept. ‘Substantively’ means at least one of the following:

- The tool is named or described as a primary feature in the record’s tool_name or tool_description fields, not as a passing mention.
- The tool’s mechanism is detailed in the record (parameters, enforcement, deployment specifics), not merely listed alongside other tools.
- The study tests, observes, or audits the tool: the tool is part of what the study reports on, not background context.

A trace is excluded if:

- The tool is mentioned only as a contextual reference (e.g., ‘alongside other RG features...’).
- The mechanism described is materially different from the Framework tool concept (e.g., a study testing deposit limits without any periodicity is not a trace for the daily, weekly, or monthly variants specifically).
- The study’s primary tool is something else and the Framework tool is incidental to the study’s design.

This rule is applied uniformly across all 210 tools.

3.12.2 Multi-tool studies

Where a single Stage 1 record substantively examines multiple Framework tools (for example, policy audits covering daily, weekly, and monthly deposit limits within the same study), the record is listed as a Taxonomy Source for each Framework tool examined. This means a small number of records appear in the Taxonomy Source field for several Framework tools.

3.12.3 Cross-code drift

Stage 1 categorised tool records by their primary tool code. Framework tool boundaries cut across Stage 1 codes for a substantial share of traces, so some studies are filed under one code in Stage 1 but substantively examine a Framework tool in a different code. Of the 602 traces recorded in the Taxonomy Source column, 117 (19.4%) link a record to a Framework tool whose Category differs from every component of the record’s Stage 1 tool code, and 93 (15.4%) link it to a Framework tool in a different Domain from the one the record was filed under (Cross_Category_Drift analysis; the broader any-difference count, which also includes compound Stage 1 codes decomposed to a single Framework Category, is 191, 31.7%). Examples:

- Donation/charity features filed under STRUCT-OTHER in Stage 1 but mapping to LIMIT-WIN tools in the Framework.
- Smart-card pre-commitment features filed under LIMIT-DEP in Stage 1 but mapping to STRUCT-PAY tools in the Framework.
- Deposit-limit promotional emails filed under CONTACT-PROACTIVE in Stage 1 but mapping to MSG-PUSH tools in the Framework.

Where such drift occurs, the trace is recorded under the Framework tool the study examines, regardless of the Stage 1 tool code.

3.12.4 Stage and modality variations

Where a study tests a tool concept at a different stage or in a different delivery modality than the Framework specifies (for example, a risk-triggered message tested in-play but the Framework tool specifies pre-login deployment), the record is included as a trace if the consolidated tool concept matches. The stage or modality variation is recorded as part of the tool's design parameters; the underlying tool concept is what the substantive trace rule evaluates.

3.12.5 Self-help framework domains

Several Stage 1 records (e.g., Rodda 2017's six-domain self-change framework, Bagot 2021's fifteen-category strategy taxonomy) catalogue many self-regulatory strategies in factor-analytic or domain-level groupings. Such records are included as a Taxonomy Source for a Framework tool only where the framework's domain or factor explicitly corresponds to the Framework tool concept (e.g., 'self-monitoring domain' → Framework T144 Self-Monitoring Journaling). Mere inclusion in a long catalogue of strategies without domain-level correspondence is not sufficient for a substantive trace.

3.12.6 Coverage

All 410 Stage 1 records are referenced at least once across the Framework's Taxonomy Source column. This is not a target outcome but a finding of the consolidation work: every record from Stage 1 substantively examines at least one tool concept that exists in the Framework. Where a record was difficult to place (typically because it tested a tool concept the Framework did not initially capture), the closest matching tool was identified and the record placed there with a note in the working log documenting any scope-stretch.

3.13 Notes

Purpose	Free-text field for tool-specific commentary that does not belong in any other column.
Format	Free text. May include design rationale, deployment caveats, links to related tools, or audit notes.
Decision rules	Used sparingly and only where informative. Notes do not affect any other column's coding. Notes may be used to flag where a tool's name or description was changed during consolidation, where the underlying evidence is contested, or where the tool is conceptually adjacent to another Framework tool.

4. Reference sheets

4.1 Domain Summary sheet

The Domain Summary sheet provides navigation between the fifteen domains and their constituent tool codes. Columns:

Domain	Domain label as used in the main RG Intervention Framework sheet (column 2).
Tool Codes	Comma-separated list of tool codes that fall within the domain. The same codes appear in column 3 of the main sheet for tools belonging to this domain.
Description	Short description of the domain's scope and the type of tools it contains.

The Domain Summary sheet does not include counts.

4.2 Tool Code Reference sheet

The Tool Code Reference sheet provides decoding for the abbreviated tool codes used in the main sheet. Columns:

Code	Hyphenated uppercase tool code (e.g., LIMIT-DEP, MSG-POP).
Code Name	Human-readable name of the tool code (e.g., ‘Deposit Limit’, ‘Pop-Up Message’).
Domain	Parent domain to which the tool code belongs.

The Tool Code Reference sheet does not include counts.

5. Versioning and updates

This codebook documents version 1.0 of the RG Intervention Framework. Future versions of the Framework may add new tools, refine controlled vocabularies, or update the Taxonomy Source mappings as new evidence becomes available.

5.1 Tool ID stability

Tool IDs are stable across versions. New tools added in a future version receive the next available ID; existing tools retain their original IDs. IDs are never reassigned or reused.

5.2 Controlled vocabulary stability

Controlled vocabularies (Target Population, Delivery Context, Governance, Personalisation, Trigger, Evidence Status) are intended to be stable. Changes to a controlled vocabulary trigger a major version increment. Additions of new values to an existing vocabulary trigger a minor version increment, with the previous values retained.

5.3 Taxonomy Source updates

Where new evidence from outside the original Stage 1 review is identified (for example, through forward-citation tracking or supplementary searches), the Taxonomy Source field may be updated to include additional study_ids. Updates of this kind do not change Tool IDs or Framework structure.

5.4 Companion file

This codebook is the canonical reference for the Framework’s structure and rules. Its companion file is 04.1 (04_1_RG_Intervention_Framework.xlsx).

5.5 Codebook revision history

Codebook revision 1.1 (September 2026): Section 3.12.3 revised to report the cross-code and cross-domain trace counts (117 of 602 and 93 of 602) in place of the earlier description of drift as occasional. The Framework file (04.1), Tool IDs, and controlled vocabularies are unchanged.

6. Stage 1 to Stage 2 Consolidation Summary

This section summarises the consolidation of the 410 Stage 1 tool records into the 210 framework tools, including trace coverage and reduction ratios. It is referenced from the main manuscript as Appendix 04.2.

Metric	Value
Tool records (taxonomy total)	410
Tool records traced into the framework	410
Coverage of tool records	100.0%
Framework tools (total)	210
Framework tools with one or more Stage 1	169 (80.5%)

traces	
Framework tools with no Stage 1 traces	41 (19.5%)
Total trace pairs (Stage 1 to Stage 2)	602
Reduction ratio across the full framework	1.95 to 1
Reduction ratio across the 176 evidence-derived framework tools	2.33 to 1

Note. A trace pair is a (tool record, framework tool) link recorded in the Taxonomy Source attribute. The 41 framework tools without a Stage 1 trace are the 34 Extrapolated framework tools (no Taxonomy Source by design), 6 Theoretically derived framework tools without a substantive trace, and 1 Policy or structural framework tool documented without empirical evaluation. The 176 evidence-derived framework tools comprise the Empirically tested, Theoretically derived, and Policy or structural categories (143 + 13 + 20). Relocated from the main manuscript.